

In the modern restaurant industry, delivering exceptional dining experiences requires more than just good cuisine. Restaurant Management Systems have emerged as the secret ingredient for efficient operations and customer satisfaction.

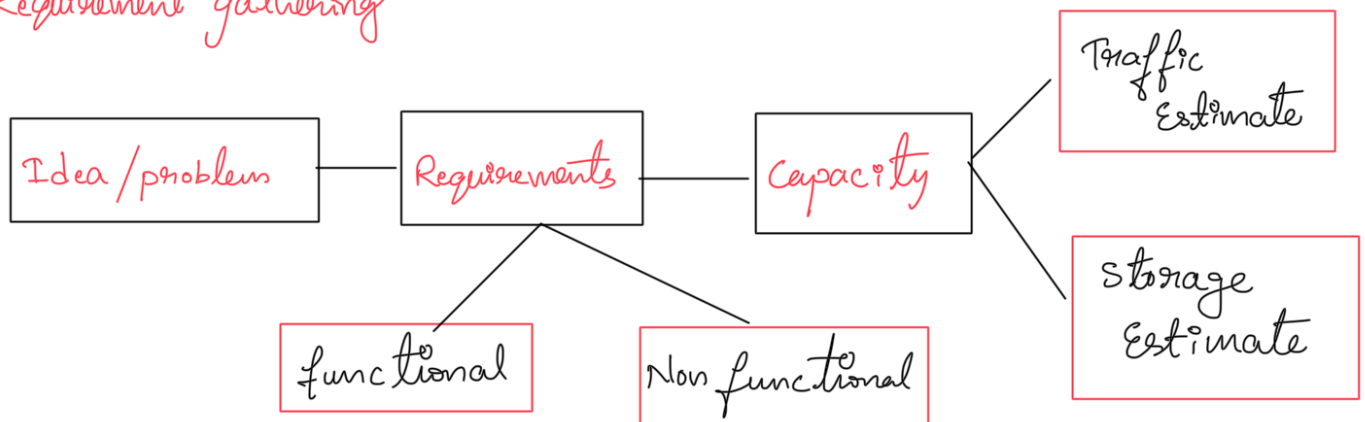
They can seamlessly get menus, explore ratings, and conveniently place orders online, with the added flexibility of choosing between dine-in or delivery to any location within the specified range.

SDLC Phases

- Requirement Gathering and Analysis:
- System Design:
- Implementation and Testing
- Deployment and Maintenance:

By following SDLC phases mentioned below, we make sure that every part of our restaurant management system—from planning what it needed to do to keeping it working well—was carefully thought out and done right.

Requirement gathering



Functional Requirements for Restaurant Management System

1. Allow customers to search restaurants, view menus, place orders.
2. Restaurant staff can manage orders, update menus.
3. Customers can view ratings of the restaurants.
4. Customers can either dine in or ask for delivery.
5. Kitchen staff can view and manage food orders.
6. Delivery drivers can view assigned delivery orders.
7. Customers can track their orders.

8. Customers can choose payment methods.

Non-functional Requirements for Restaurant Management System

1. **Scalability:** It ensures the system remains responsive during peak demand, adapts to the addition of new features, and handles fluctuations in traffic.
2. **Reliability:** Reliability is paramount for customer trust and business continuity. Users must have confidence in the system's consistent and accurate order processing, and downtime should be minimized for restaurant owners and delivery agents relying on the platform.
3. **High Availability:** A reliable and always-accessible system enhances customer trust, supports efficient restaurant operations, and ultimately contributes to the success of the entire restaurant management ecosystem.
4. **Strong consistency:** Any changes in price, menu, location of user, payment details should reflect quickly.

Capacity Assumptions

To estimate the scale of the system and to get the idea about the storage requirements, we have to make some assumptions about the stored data and traffic. Following are some assumptions for the given design.

- 10 million restaurants
- Daily active users = 100 million
- Active users per second = 1160 users/sec (approx.)
- Average Orders per minute = 250,000/min
- Average Orders per second = 4200 orders/second (approx.)
- Peak orders = 2 * Average orders per minute = 500,000 orders/minute

Let's see the storage estimation for Restaurant Management System

Let's assume on average

- Each customer row in DB is 1 KB (with address, profile info etc.)
- Each restaurant = 2 MB (details, menus, photos etc.)
- Each order = 1 KB.

Total storage needed:

- 100 million customers x 1KB per user = 100 GB
- 10 million restaurants x 2 MB per restaurant = 20 TB

- 500,000 orders per minute = 72 million orders per day (approx.)
- 72 million orders per day x 1 KB per order = 72 GB per day

Assuming we maintain orders for last 3 years:

72 GB per day x 365 days x 3 years = 77TB (approx.)